

MCA SEM II Subject: Software Project Management

Unit 6: Introduction of Agile Tools

GitHub:

GitHub is a web-based version-control and collaboration platform for software developers. Microsoft, the biggest single contributor to GitHub. GitHub, which is delivered through a software-as-a-service (SaaS) business model, was started in 2008 and was founded on Git, an open source code management system .

Git is used to store the source code for a project and track the complete history of all changes to that code. It allows developers to collaborate on a project more effectively by providing tools for managing possibly conflicting changes from multiple developers. GitHub allows developers to change, adapt and improve software from its public repositories for free, but it charges for private repositories, offering various paid plans. Each public or private repository contains all of a project's files.

GitHub facilitates **social coding** by providing a web interface to the Git code repository and management tools for collaboration. GitHub can be thought of as a serious social networking site for software developers. Members can follow each other, rate each other's work, receive updates for specific projects **and communicate publicly or privately.**

Three important terms used by developers in GitHub are ***Fork, Pull Request and Merge.***

A *fork*, also known as a *branch*, is simply a repository that has been copied from one member's account to another member's account. Forks and branches allow a developer to make modifications without affecting the original code.

If the developer would like to share the modifications, she can send a *pull request* to the owner of the original repository. If, after reviewing the modifications, the original owner would like to pull the modifications into the repository, she can accept the modifications and *merge* them with the original repository. Because GitHub is so intuitive to use and its version-control tools are so useful for collaboration, nonprogrammers have also begun to use GitHub to work on document-based and multimedia projects.

GitHub products and features

GitHub offers an on-premises version. GitHub Enterprise supports integrated development environments.

Other products and features of note include:

- **GitHub Gist** allows GitHub users to share pieces of code or other notes.
- **GitHub Flow** is a lightweight workflow for regularly updated deployments.
- **GitHub Pages** are static webpages to host a project, pulling information directly from an individual's or organization's GitHub repository.
- **GitHub Desktop** enables users to access GitHub from Windows or Mac desktops, rather than going to GitHub's website.

- **GitHub Student Developer Pack** is a free offering of developer tools that is limited to students, and includes cloud resources, programming tools and support, and GitHub access.

Introduction to Agile Tools

Agile tools are project management tools designed to support agile methodology.

Though there is a huge list of agile project management tools, few tools are described as below:

1. Active Collab

This is a powerful management tool that helps in managing from juggling tasks to tracking time as well as generating bills. It provides a system-wide calendar that helps to comprehend the tasks assigning to each team member. Active collab has a strong documentation system that supports a collaborative writing tool that each member can take part in the documentation.

2. Jira

Jira is one of the most well-known and proven products used by many companies around the globe. It is highly widespread and customizable on the basis of requirement. It has a customizable Scrum Board, which can be used to examine all the tasks in the sprint, and even it shows the accurate picture of the sprint with the help of a burnout chart, sprint report, velocity chart, epic report, a cumulative flow

diagram, etc. It provides custom filters using JQL (Jira Query Language). In addition, flexible Kanban boards deliver the maximum output with minimum effort.

Jira, a customizable project management tool, is also not out of flaws. The biggest flaw is its complex nature to use. Just only for setting up, it requires developers and takes a long time to learn its function properly. On the other hand, Jira is a highly expensive project management tool.

3. Lean kit

The lean kit uses cards to show work items and the current status of the on-going project. It is flexible enough to give space to the team members to post all the task details, user stories as virtual notes or cards. In addition, multiple team members can get the chance to work together in separate spaces with coordinating with each other. Basically, this software system allows replicating the conference room from where all tasks related to the project begins.

4. Pivotal Tracker

Pivotal tracker bears a group of tools to aid in agile project methodology, and it is ideal for web and mobile developers. The group of tools includes Whiteboard for project-related discussion, Project monitor for showing the status and sprout. The main feature of this software application is keeping a record of all the tasks as stories from inception to completion of the project. It provides tools to monitor each day's task, and the team members can point out the complexity with notes.

5. VersionOne

Version one is ideal for IT professionals and tech-savvy designers. The tools of this software embrace methodologies like Kanban, DAD, LeSS and the Hybrid

approach. Version one provides space for collaboration and stable communication among multiple team members who have engaged in the development process of the enterprise. So every team member can get the chance to plan initiatives and documentation accordingly.

6. Planbox

Planbox provides features to collaborate with multiple teams work to attain a specific goal.

Planbox evaluates the progress of the burndown chart, and it also assimilates customer feedback, bug report for the utilization of users.

7. Axosoft

Axosoft is scrum management software which helps to examine every step of the development process and make graph report accordingly. The tools of Axosoft attribute progress differently. The release planner, one of the tools of Axosoft, gives a tabular view of various tasks, bugs, and user stories. The developers organize different entries, and the burndown chart represents the performance report graphically. One of the important features of Axosoft which makes it different from others is the customer portal.

8. Taiga

Taiga is an ideal software application for content developers of various verticals. It supports Scrum and Kanban methodologies with the ability to create product backlogs with the sprints and user stories. It provides video call to team members,

9. Hygger

Hugger is a perfect combination of project management tools providing easy solutions for each task, including establishing project goals and divide them into various tasks, distributing the tasks on the Kanban board and fixing prioritization through the Value/Effort matrix collaborating and communicating with the team members. It provides almost all features what is required for the completion of any kind of project.